SENTIMENT & SARCASM DETECTION

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INTRODUCTION

Sentiment analysis is an important NLP task used to understand opinions and emotions in text, especially on social media. However, traditional sentiment analysis models often fail to correctly interpret sarcastic expressions, which can reverse the intended meaning of a sentence and lead to incorrect sentiment classification. Therefore, handling sarcasm is crucial for improving the accuracy of sentiment analysis.

PROJECT OBJECTIVE

The project aims to develop a sentiment analysis model to classify text as positive or negative, and a sarcasm detection model to identify sarcastic expressions in text. By combining sentiment analysis with sarcasm detection, the project improves the understanding of the true meaning of text. The performance of both models is evaluated using standard classification metrics to ensure accurate and reliable results.

DATASETS USED

datasets from Kaggle:

- Sarcasm Data: Balanced sarcasm dataset with 100k comments labeled sarcastic or not.
- Sentiment140 Dataset : Sentiment140 dataset with 1.6M tweets labeled positive or negative.

DATA PREPROCESSING

- Clean text by removing URLs, mentions, punctuation, and converting to lowercase.
- Use TF-IDF vectorization to convert text into numerical features for model training.

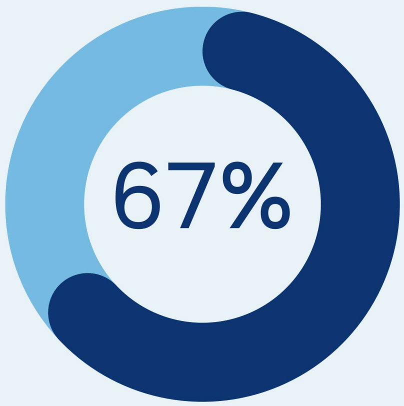
MODEL TRAINING

Train-test split: 80% training, 20% testing

Train separate models for:

- Sentiment Analysis
- Sarcasm Detection

Model trained using **TF-IDF features**



MODELS USED

Logistic Regression (Sentiment) , (Sarcasm)

Support vector machine SVM (Sentiment) , (Sarcasm)

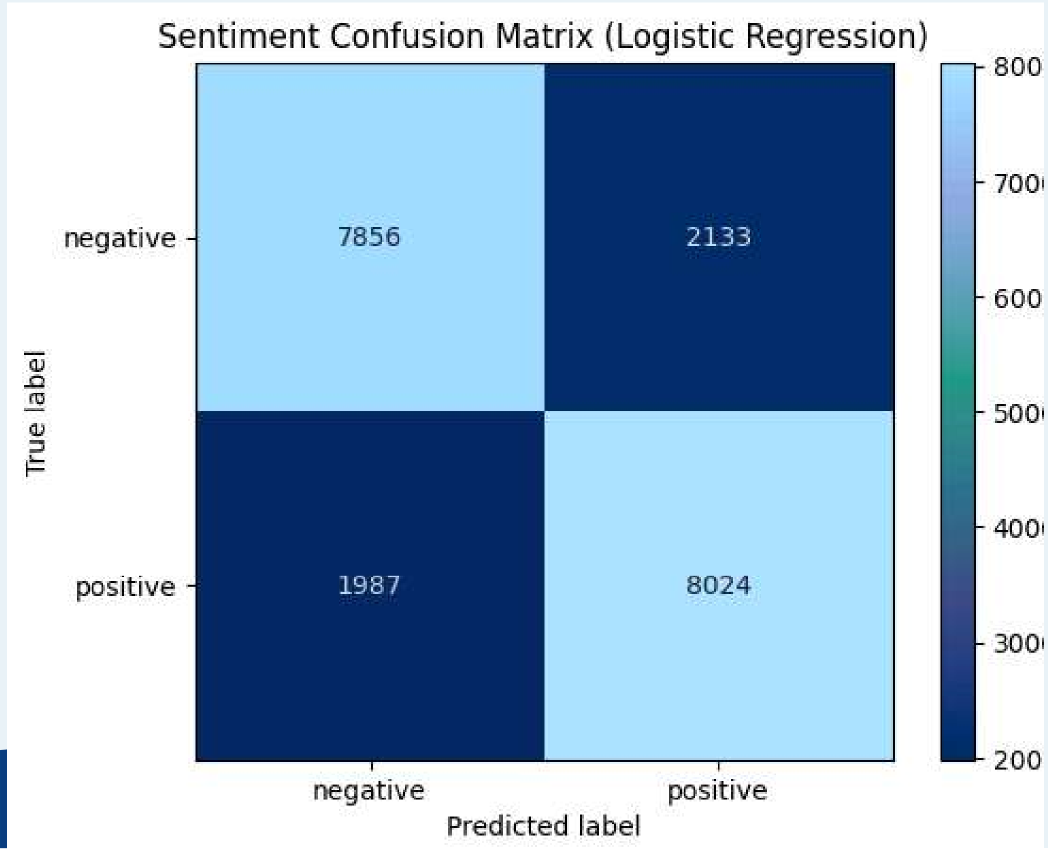
Ensemble logic **(Rule-based sentiment adjustment):** If text is detected as sarcastic, adjust sentiment accordingly

MODEL PERFORMANCE

Sentiment Analysis

Model	Accuracy	F1 -S -neg	F1-S-pos
Logistic Regression	0.794	0.79	0.80
SVM	0.7808	0.78	0.78

CONFUSION MATRIX (LOGISTIC REGRESSION)

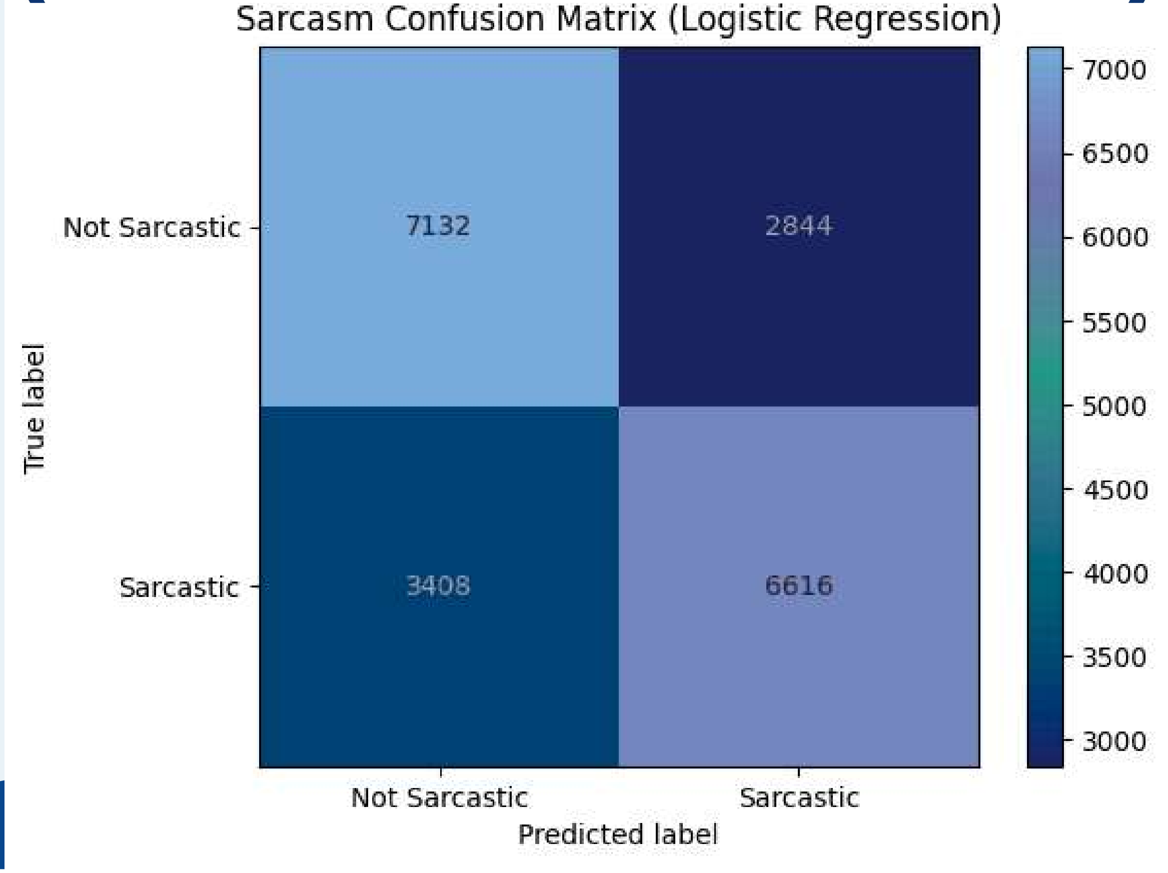


MODEL PERFORMANCE

Sarcasm Detection

Model	Accuracy	F1 -S -neg	F1-S-pos
Logistic Regression	0.6874	0.70	0.68
SVM	0.666	0.67	0.66

CONFUSION MATRIX (LOGISTIC REGRESSION)



HOW PREDICTIONS WORK

EXAMPLE PREDICTIONS

text	Sentiment	Sarcasm	Adjusted Sentiment
I just love waiting in traffic for hours	Positive	Sarcastic	Negative
This phone is amazing and works perfectly	Positive	Not sarcasm	Positive

THANK YOU